```
> restart;#SchwarAxR;
> with(PDEtools):
> with(Physics):
> with(LinearAlgebra):
> Setup(mathematicalnotation = true)
```

$$[mathematicalnotation = true] \tag{1}$$

```
> Setup(coordinatesystems = cartesian)
```

Systems of spacetime coordinates are: $\{X = (x, y, z, t)\}$

$$[coordinatesystems = \{X\}] \tag{2}$$

```
> g_[];
```

$$g_{\mu,\nu} = \begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{3}$$

```
> ds2 := r^2 dtheta^2 + r^2 sin(θ)^2 dphi^2 - r·c^2·dr^2/(2 m G - r·c^2) - (r c^2 - 2 m G)/r · dt^2 ;
  # The Schwarzchild metric
```

$$ds2 := r^2\, d\theta^2 + r^2 \sin(\theta)^2\, d\phi^2 - \frac{r\, c^2\, dr^2}{-r\, c^2 + 2\, m\, G} - \frac{(r\, c^2 - 2\, m\, G)\, dt^2}{r} \tag{4}$$

```
> Setup(coordinates = spherical, metric = ds2);
```

Systems of spacetime coordinates are: $\{X = (r, \theta, \phi, t)\}$

Coordinates: $[r, \theta, \phi, t]$. *Signature:* $(- - - +)$

$$g_{\mu,\nu} = \begin{bmatrix} -\dfrac{r\, c^2}{-r\, c^2 + 2\, m\, G} & 0 & 0 & 0 \\ 0 & r^2 & 0 & 0 \\ 0 & 0 & r^2 \sin(\theta)^2 & 0 \\ 0 & 0 & 0 & \dfrac{-r\, c^2 + 2\, m\, G}{r} \end{bmatrix}$$

Setting lowercaselatin_is letters to represent space indices

$$\left[coordinatesystems = \{X\}, metric = \left\{(1,1) = -\frac{r\, c^2}{-r\, c^2 + 2\, m\, G}, (2,2) = r^2, (3,3) = r^2 \sin(\theta)^2, (4,4) = \right.\right. \tag{5}$$

$$-\frac{r\,c^2-2\,m\,G}{r}\Big\},\ spaceindices = lowercaselatin_is\Big]$$

```
> Ricci[scalarsdefinition];
```

$$\Phi_{00}=-\frac{R^{\mu,\nu}\,l_\mu\,l_\nu}{2},\ \Phi_{01}=-\frac{R^{\mu,\nu}\,l_\mu\,m_\nu}{2},\ \Phi_{02}=-\frac{R^{\mu,\nu}\,m_\mu\,m_\nu}{2},\ \Phi_{11}=-\frac{R^{\mu,\nu}\left(l_\mu\,n_\nu+m_\mu\,\overline{m}_\nu\right)}{4},\ \Phi_{12}= \tag{6}$$

$$-\frac{R^{\mu,\nu}\,m_\mu\,n_\nu}{2},\ \Phi_{22}=-\frac{R^{\mu,\nu}\,n_\mu\,n_\nu}{2},\ \Lambda=\frac{R^{\mu}}{24}$$

```
> Ricci[scalars];
```

$$\Phi_{00}=0,\ \Phi_{01}=0,\ \Phi_{02}=0,\ \Phi_{11}=0,\ \Phi_{12}=0,\ \Phi_{22}=0,\ \Lambda=0 \tag{7}$$

```
> Ricci[scalar];
```

$$0 \tag{8}$$

```
> TR := r = (-ρ·c^2 + m·G)/(m·G)
    # Rs = 2 G·m / c^2  the Schwarschields radius so we have the function r = 1
```

$$-\frac{2\cdot\rho}{Rs}$$

$$TR := r=\frac{-\rho\,c^2+m\,G}{m\,G} \tag{9}$$

```
> subs(r = rho, TransformCoordinates(TR, g_[mu, nu], [rho, theta, phi, t]));
    #,setmetric
```

$$\begin{bmatrix} -\dfrac{c^6\left(-\rho\,c^2+m\,G\right)}{m^2\,G^2\left(c^4\,\rho+2\,G^2\,m^2-c^2\,m\,G\right)} & 0 & 0 & \cdots \\ 0 & \dfrac{\left(-\rho\,c^2+m\,G\right)^2}{m^2\,G^2} & 0 & \cdots \\ 0 & 0 & \dfrac{\left(-\rho\,c^2+m\,G\right)^2\sin(\theta)^2}{m^2\,G^2} & \cdots \\ 0 & 0 & 0 & \cdots \end{bmatrix} \tag{10}$$

```
> TransformCoordinates(TR, g_[mu, nu], [rho, theta, phi, t], output
    = line_element)
```

$$-\frac{1}{m^2\,G^2\left(c^4\,\rho+2\,G^2\,m^2-c^2\,m\,G\right)}\Big(\Big(\left(\mathrm{d}(\rho)\right)^2c^6-\left(c^4\,\rho+2\,G^2\,m^2-c^2\,m\,G\right)\left(-\rho\,c^2+m\,G\right)\left(\sin(\theta)^2\left(\mathrm{d}(\phi)\right)^2+\left(\mathrm{d}(\theta)\right)^2\right)\Big)\left(-\rho\,c^2+m\,G\right)\Big)+\frac{\left(c^4\,\rho+2\,G^2\,m^2-c^2\,m\,G\right)\left(\mathrm{d}(t)\right)^2}{-\rho\,c^2+m\,G} \tag{11}$$

```
> subs(ρ = r, TransformCoordinates(TR, g_[mu, nu], [rho, theta, phi, t], output
```

```
    = line_element));
```

$$-\frac{1}{m^2 G^2 \left(c^4 r+2 m^2 G^2-G c^2 m\right)}\left(\left(\left(\mathrm{d}(r)\right)^2 c^6-\left(c^4 r+2 m^2 G^2-G c^2 m\right)\left(-r c^2+m G\right)\left(\sin(\theta)^2\left(\right.\right.\right.\right. \tag{12}$$

$$\left.\left.\left.\left.\mathrm{d}(\phi)\right)^2+\left(\mathrm{d}(\theta)\right)^2\right)\right)\left(-r c^2+m G\right)\right)+\frac{\left(c^4 r+2 m^2 G^2-G c^2 m\right)\left(\mathrm{d}(t)\right)^2}{-r c^2+m G}$$

```
> ds21 := (-r· c^2 + m G)^2 /(m^2 G^2) ·dtheta^2 + (-r· c^2 + m G)^2 /(m^2 G^2) · sin(θ)^2 dphi^2
    - c^6 (-r· c^2 + m G) /(m^2 G^2 (c^4 r + 2 m^2 G^2 - m G c^2)) ·dr^2 + (c^4 r + 2 m^2 G^2 - m G c^2)/(-r c^2 + m G) ·dt^2;
```

$$ds21 := \frac{\left(-r c^2+m G\right)^2 d\theta^2}{m^2 G^2}+\frac{\left(-r c^2+m G\right)^2 \sin(\theta)^2 d\phi^2}{m^2 G^2}-\frac{c^6\left(-r c^2+m G\right) dr^2}{m^2 G^2\left(c^4 r+2 m^2 G^2-m G c^2\right)} \tag{13}$$

$$+\frac{\left(c^4 r+2 m^2 G^2-m G c^2\right) dt^2}{-r c^2+m G}$$

```
> Setup(coordinates = spherical, metric = ds21);
```

Systems of spacetime coordinates are: $\{X=(r, \theta, \phi, t)\}$

Coordinates: $[r, \theta, \phi, t]$. *Signature:* $(-\,-\,-\,+)$

$g_{\mu,\nu} =$

$$\begin{bmatrix} -\frac{c^6\left(-r c^2+m G\right)}{m^2 G^2\left(c^4 r+2 m^2 G^2-m G c^2\right)} & 0 & 0 & \cdots \\ 0 & \frac{\left(-r c^2+m G\right)^2}{m^2 G^2} & 0 & \cdots \\ 0 & 0 & \frac{\left(-r c^2+m G\right)^2 \sin(\theta)^2}{m^2 G^2} & \cdots \\ 0 & 0 & 0 & \cdots \end{bmatrix}$$

$$\left[coordinatesystems=\{X\}, metric=\left\{(1,1)=-\frac{c^6\left(-r c^2+m G\right)}{m^2 G^2\left(c^4 r+2 m^2 G^2-m G c^2\right)}, (2,2)\right.\right. \tag{14}$$

$$\left.\left.=\frac{\left(-r c^2+m G\right)^2}{m^2 G^2}, (3,3)=\frac{\left(-r c^2+m G\right)^2 \sin(\theta)^2}{m^2 G^2}, (4,4)=\frac{c^4 r+2 m^2 G^2-m G c^2}{-r c^2+m G}\right\}\right]$$

```
> Ricci[scalarsdefinition];
```

$$\Phi_{00} = -\frac{R^{\mu,\nu}\, l_\mu\, l_\nu}{2},\ \Phi_{01} = -\frac{R^{\mu,\nu}\, l_\mu\, m_\nu}{2},\ \Phi_{02} = -\frac{R^{\mu,\nu}\, m_\mu\, m_\nu}{2},\ \Phi_{11} = -\frac{R^{\mu,\nu}\left(l_\mu\, n_\nu + m_\mu\, \overline{m}_\nu\right)}{4},\ \Phi_{12} = \tag{15}$$

$$-\frac{R^{\mu,\nu}\, m_\mu\, n_\nu}{2},\ \Phi_{22} = -\frac{R^{\mu,\nu}\, n_\mu\, n_\nu}{2},\ \Lambda = \frac{R^{\mu}}{24}$$

```
> Ricci[scalars];
```

$$\Phi_{00} = 0,\ \Phi_{01} = 0,\ \Phi_{02} = 0,\ \Phi_{11} = 0,\ \Phi_{12} = 0,\ \Phi_{22} = 0,\ \Lambda = 0 \tag{16}$$

```
> Ricci[scalar];
```

$$0 \tag{17}$$

```
> EnergyMomentum[definition] ; EnergyMomentum[ ]
```

$$T_{\mu,\nu} = \frac{G_{\mu,\nu}}{8\,\pi}$$

$$T_{\mu,\nu} = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \tag{18}$$

```
> f(r, t) := (-r· c^2 + m G)^2/(m^2 G^2)·dtheta^2 + (-r· c^2 + m G)^2/(m^2 G^2) ·sin(θ)^2 dphi^2
  - c^6 (-r· c^2 + m G)/(m^2 G^2 (c^4 r + 2 m^2 G^2 - m G c^2))·dr^2 + (c^4 r + 2 m^2 G^2 - m G c^2)/(-r c^2 + m G)· dt^2 ;
```

$$f := (r,t) \mapsto \left(-r\,c^2 + m\,G\right)^2 \frac{1}{m^2\,G^2}\,d\theta^2 + \left(-r\,c^2 + m\,G\right)^2 \frac{1}{m^2\,G^2}\,\sin(\theta)^2\,d\phi^2 - c^6\left(-r\,c^2 + m\,G\right)\frac{1}{m^2\,G^2\left(c^4\,r + 2\,m^2\,G^2 - m\,G\,c^2\right)}\,dr^2 + \left(c^4\,r + 2\,m^2\,G^2 - m\,G\,c^2\right)\frac{1}{-r\,c^2 + m\,G}\,dt^2 \tag{19}$$

```
> change_of_vars := {ρ = m·G(1 - r)/c^2};
  # The transformation inverse of relation (9) ρ=Rsch(1 - r)/2; RSchwar
  = 2·m·G/c^2 ;
```

$$change_of_vars := \left\{\rho = \frac{m\,G(1-r)}{c^2}\right\} \tag{20}$$

```
> tmp := dchange(change_of_vars, [f(ρ, t)], [r], params = m, G, c, simp_proc);
  # and we simplify expresion
```

$$tmp := \left[\frac{\left(-m\,G(1-r) + m\,G\right)^2 d\theta^2}{m^2\,G^2} + \frac{\left(-m\,G(1-r) + m\,G\right)^2 \sin(\theta)^2\,d\phi^2}{m^2\,G^2} - \frac{c^6\left(-m\,G(1-r) + m\,G\right)dr^2}{m^2\,G^2\left(c^2\,m\,G(1-r) + 2\,m^2\,G^2 - m\,G\,c^2\right)} + \frac{\left(c^2\,m\,G(1-r) + 2\,m^2\,G^2 - m\,G\,c^2\right)dt^2}{-m\,G(1-r) + m\,G}\right] \tag{21}$$

```
> simplify((G-G+r·G)^2/G^2); simplify(-c^6·r·m·G/(m^3·G^3·(2 G m-c^2+c^2-r·c^2))·dr^2);
  simplify((2 G^2 m-G c^2+G·c^2-r·G· c^2)/(G·r)·dt^2);
```

$$\frac{r^2}{}$$

$$-\frac{c^6\, r\, dr^2}{m^2\, G^2\, \left(-r\, c^2+2\, m\, G\right)}$$

$$\frac{\left(-r\, c^2+2\, m\, G\right)\, dt^2}{r} \tag{22}$$

```
> ds22 := r^2·dtheta^2+r^2·sin(θ)^2·dphi^2-c^6·r/(G^2 m^2 (-r c^2+2 m G))·dr^2
  +(-r c^2+2 m G)/r·dt^2 ;
```

$$ds22 := r^2\, d\theta^2+r^2\, \sin(\theta)^2\, d\phi^2-\frac{c^6\, r\, dr^2}{m^2\, G^2\, \left(-r\, c^2+2\, m\, G\right)}+\frac{\left(-r\, c^2+2\, m\, G\right)\, dt^2}{r} \tag{23}$$

```
> Setup(coordinates=spherical, metric=ds22);
```

Systems of spacetime coordinates are: $\{X=(r, \theta, \phi, t)\}$

Coordinates: $[r, \theta, \phi, t]$. *Signature:* $(-\,-\,-\,+)$

$$g_{\mu,\nu}=\begin{bmatrix} -\frac{c^6\, r}{m^2\, G^2\, \left(-r\, c^2+2\, m\, G\right)} & 0 & 0 & 0 \\ 0 & r^2 & 0 & 0 \\ 0 & 0 & r^2\, \sin(\theta)^2 & 0 \\ 0 & 0 & 0 & \frac{-r\, c^2+2\, m\, G}{r} \end{bmatrix}$$

$$\left[coordinatesystems=\{X\}, metric=\left\{(1,1)=-\frac{c^6\, r}{m^2\, G^2\, \left(-r\, c^2+2\, m\, G\right)}, (2,2)=r^2, (3,3)=r^2\, \sin(\theta)^2, (4,4)=\frac{-r\, c^2+2\, m\, G}{r}\right\}\right] \tag{24}$$

```
> Ricci[scalarsdefinition];
```

$$\Phi_{00}=-\frac{R^{\mu,\nu}\, l_\mu\, l_\nu}{2}, \Phi_{01}=-\frac{R^{\mu,\nu}\, l_\mu\, m_\nu}{2}, \Phi_{02}=-\frac{R^{\mu,\nu}\, m_\mu\, m_\nu}{2}, \Phi_{11}=-\frac{R^{\mu,\nu}\, \left(l_\mu\, n_\nu+m_\mu\, \overline{m}_\nu\right)}{4}, \Phi_{12}= \tag{25}$$

$$-\frac{R^{\mu,\nu}\, m_\mu\, n_\nu}{2},\ \Phi_{22} = -\frac{R^{\mu,\nu}\, n_\mu\, n_\nu}{2},\ \Lambda = \frac{R^{\mu}_{\ \mu}}{24}$$

```
> Ricci[scalars];
```

$$\Phi_{00} = 0,\ \Phi_{01} = 0,\ \Phi_{02} = 0,\ \Phi_{11} = \frac{-m^2 G^2 + c^4}{4 r^2 c^4},\ \Phi_{12} = 0,\ \Phi_{22} = 0,\ \Lambda = \frac{-m^2 G^2 + c^4}{12 r^2 c^4} \tag{26}$$

```
> Ricci[scalar];
```

$$-\frac{2\left(m^2 G^2 - c^4\right)}{r^2 c^4} \tag{27}$$

```
> g_[ ]; Λ := (-m^2 G^2 + c^4)/(12 r^2 c^4); Ricci[~μ, ν, matrix];
```

$$g_{\mu,\nu} = \begin{bmatrix} -\frac{c^6 r}{m^2 G^2\left(-r c^2 + 2 m G\right)} & 0 & 0 & 0 \\ 0 & r^2 & 0 & 0 \\ 0 & 0 & r^2 \sin(\theta)^2 & 0 \\ 0 & 0 & 0 & \frac{-r c^2 + 2 m G}{r} \end{bmatrix}$$

$$\Lambda := \frac{-m^2 G^2 + c^4}{12 r^2 c^4}$$

$$R^{\mu}_{\ \nu} = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & \frac{-m^2 G^2 + c^4}{r^2 c^4} & 0 & 0 \\ 0 & 0 & \frac{-m^2 G^2 + c^4}{r^2 c^4} & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \tag{28}$$

```
>
```

```
> ds22 := r^2·dtheta^2 + r^2 ·sin(θ)^2 ·dphi^2 - c^6 r/(G^2 m^2 (-r c^2 + 2 m G))·dr^2 + (-r c^2 + 2 m G)/r
    ·dt^2 ;  # We simplify the metric equation (22)
```

$$ds22 := r^2\, d\theta^2 + r^2 \sin(\theta)^2\, d\phi^2 - \frac{c^6 r\, dr^2}{m^2 G^2\left(-r c^2 + 2 m G\right)} + \frac{\left(-r c^2 + 2 m G\right) dt^2}{r} \tag{29}$$

```
> (1/2 ·g_[ ]·Ricci[scalar] - Ricci[~μ, ν, matrix] + Λ·g_[ ]);
```

$$-\frac{g_{\mu,\nu}\left(m^2 G^2 - c^4\right)}{r^2 c^4} - R^{\mu}_{\ \nu} + \frac{\left(-m^2 G^2 + c^4\right) g_{\mu,\nu}}{12 r^2 c^4} = \tag{30}$$

$$\begin{bmatrix} \frac{\left(m^2 G^2 - c^4\right) c^2}{r m^2 G^2 \left(-r c^2 + 2 m G\right)} - \frac{\left(-m^2 G^2 + c^4\right) c^2}{12 r m^2 G^2 \left(-r c^2 + 2 m G\right)} & \cdots \\ 0 & -\frac{m^2 G^2 - c^4}{c^4} - \cdots \\ 0 & \cdots \\ 0 & \cdots \end{bmatrix}$$

$$= \frac{1}{2} \cdot \frac{2\left(-m^2 G^2 + c^4\right)}{r^2 c^4} \cdot \begin{bmatrix} -\frac{c^6 r}{m^2 G^2 \left(-r c^2 + 2 m G\right)} & 0 & 0 & 0 \\ 0 & r^2 & 0 & 0 \\ 0 & 0 & r^2 \sin(\theta)^2 & 0 \\ 0 & 0 & 0 & \frac{-r c^2 + 2 m G}{r} \end{bmatrix}$$

$$- \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & \frac{-m^2 G^2 + c^4}{r^2 m^2 G^2} & 0 & 0 \\ 0 & 0 & \frac{-m^2 G^2 + c^4}{r^2 m^2 G^2} \cdot \sin(\theta)^2 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} + \frac{-m^2 G^2 + c^4}{12 r^2 c^4}$$

$$\cdot \begin{bmatrix} -\frac{c^6 r}{m^2 G^2 \left(-r c^2 + 2 m G\right)} & 0 & 0 & 0 \\ 0 & r^2 & 0 & 0 \\ 0 & 0 & r^2 \sin(\theta)^2 & 0 \\ 0 & 0 & 0 & \frac{-r c^2 + 2 m G}{r} \end{bmatrix};$$

(31)

$$\begin{bmatrix} -\frac{13\left(-m^2G^2+c^4\right)c^2}{12\,r\,m^2G^2\left(-rc^2+2\,mG\right)} & 0 & \cdots \\ 0 & \frac{13\left(-m^2G^2+c^4\right)}{12\,c^4}-\frac{-m^2G^2+c^4}{r^2m^2G^2} & \cdots \\ 0 & 0 & \frac{13}{\cdots} \\ 0 & 0 & \cdots \end{bmatrix} \tag{31}$$

```
> factor( -13 (-m^2 G^2 + c^4) c^2 / (12 r m^2 G^2 (-r c^2 + 2 m G)) ); factor( 13 (-m^2 G^2 + c^4) / (12 c^4) - (-m^2 G^2 + c^4) / (r^2 m^2 G^2) );
  factor( 13 (-m^2 G^2 + c^4) (-r c^2 + 2 m G) / (12 r^3 c^4) );
```

$$\frac{13\left(mG-c^2\right)\left(mG+c^2\right)c^2}{12\,r\,m^2G^2\left(-rc^2+2\,mG\right)}$$

$$-\frac{\left(mG-c^2\right)\left(mG+c^2\right)\left(13\,G^2m^2r^2-12\,c^4\right)}{12\,c^4G^2m^2r^2}$$

$$-\frac{13\left(mG-c^2\right)\left(mG+c^2\right)\left(-rc^2+2\,mG\right)}{12\,r^3c^4} \tag{32}$$

```
>
```

```
> N := (m G - c^2) (m G + c^2) · [ [ -13/12 c^2 / (r m^2 G^2 (-r c^2 + 2 m G)), 0, 0, 0 ],
    [ 0, (13 r^2 m^2 G^2 - 12 c^4) / (12 c^4 r^2 m^2 G^2), 0, 0 ],
    [ 0, 0, (13 r^2 m^2 G^2 - 12 c^4) / (12 c^4 r^2 m^2 G^2) · sin(θ)^2, 0 ],
    [ 0, 0, 0, -13·(-r c^2 + 2 m G) / (12 r^3 c^4) ] ];
```

$N :=$ (33)

$$\begin{bmatrix} -\frac{13\left(mG-c^2\right)\left(mG+c^2\right)c^2}{12\,r\,m^2\,G^2\left(-rc^2+2\,mG\right)} & 0 & \cdots \\ 0 & \frac{\left(mG-c^2\right)\left(mG+c^2\right)\left(13\,G^2m^2r^2-12\,c^4\right)}{12\,c^4\,G^2\,m^2\,r^2} & \cdots \\ 0 & 0 & \cdots \\ 0 & 0 & \cdots \end{bmatrix}$$

```
> solve(N(1, 1) = 0);
```

Warning, solve may be ignoring assumptions on the input variables.

$$\{G=G, c=0, m=m, r=r\}, \left\{G=\frac{c^2}{m}, c=c, m=m, r=r\right\}, \left\{G=-\frac{c^2}{m}, c=c, m=m, r=r\right\} \quad (34)$$

```
> solve(N(2, 2) = 0);
```

Warning, solve may be ignoring assumptions on the input variables.

$$\left\{G=\frac{c^2}{m}, c=c, m=m, r=r\right\}, \left\{G=-\frac{c^2}{m}, c=c, m=m, r=r\right\}, \left\{G=\frac{2\sqrt{39}\,c^2}{13\,r\,m}, c=c, m=m, r=r\right\}, \left\{G=-\frac{2\sqrt{39}\,c^2}{13\,r\,m}, c=c, m=m, r=r\right\} \quad (35)$$

```
> solve(N(4, 4) = 0);
```

Warning, solve may be ignoring assumptions on the input variables.

$$\left\{G=\frac{c^2}{m}, c=c, m=m, r=r\right\}, \left\{G=-\frac{c^2}{m}, c=c, m=m, r=r\right\}, \left\{G=\frac{r\,c^2}{2\,m}, c=c, m=m, r=r\right\} \quad (36)$$

```
> N := [[0, 0, 0, 0], [0, 0, 0, 0], [0, 0, 0, 0], [0, 0, 0, 0]];
```

#For $m=\frac{c^2}{G}$ we have null matrix thus null Tensor StresEnergy

$$N := \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad (37)$$

```
> EnergyMomentum[mu, p] = Array([[0, 0, 0, 0], [0, 0, 0, 0], [0, 0, 0, 0], [0, 0, 0, 0]])
```

$$T_{\mu,p} = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \tag{38}$$

```
> restart;
> with(Physics):
> with(PDEtools):
> with(LinearAlgebra):
> Setup(mathematicalnotation = true)
```

$$[mathematicalnotation = true] \tag{39}$$

```
> Setup(coordinatesystems = cartesian)
```

Systems of spacetime coordinates are: $\{X = (x, y, z, t)\}$

$$[coordinatesystems = \{X\}] \tag{40}$$

```
> g_[];
```

$$g_{\mu,\nu} = \begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{41}$$

```
> ds22 := r^2 ·dtheta^2 + r^2 ·sin(theta)^2 ·dphi^2 - c^6 r/(m^2 G^2 (-r c^2 + 2 m G))·dr^2
  + (-r c^2 + 2 m G)/r ·dt^2 ;
```

$$ds22 := r^2\, d\theta^2 + r^2 \sin(\theta)^2\, d\phi^2 - \frac{c^6 r\, dr^2}{m^2 G^2 \left(-r c^2 + 2 m G\right)} + \frac{\left(-r c^2 + 2 m G\right) dt^2}{r} \tag{42}$$

```
> Setup(coordinates = spherical, metric = ds22);
```

Systems of spacetime coordinates are: $\{X = (r, \theta, \phi, t)\}$

Coordinates: $[r, \theta, \phi, t]$. *Signature:* $(- - - +)$

$$g_{\mu,\nu} = \begin{bmatrix} -\frac{c^6 r}{m^2 G^2 \left(-c^2 r + 2 m G\right)} & 0 & 0 & 0 \\ 0 & r^2 & 0 & 0 \\ 0 & 0 & r^2 \sin(\theta)^2 & 0 \\ 0 & 0 & 0 & \frac{-c^2 r + 2 m G}{r} \end{bmatrix}$$

Setting lowercaselatin_is letters to represent space indices

$$\left[coordinatesystems = \{X\},\ metric = \left\{(1,1) = -\frac{c^6 r}{m^2 G^2 \left(-c^2 r + 2 m G\right)},\ (2,2) = r^2,\ (3,3) = r^2 \sin(\theta)^2,\ (4,4) = \frac{-c^2 r + 2 m G}{r}\right\},\ spaceindices = lowercaselatin_is\right] \quad (43)$$

```
> simplify( -c^6 r/(m^2 G^2 (-r c^2 + 2 m G)) · dr^2 ); simplify( (-r c^2 + 2 m G)/r · dt^2 );
```

$$-\frac{c^6 r\, dr^2}{m^2 G^2 \left(-c^2 r + 2 m G\right)}$$

$$\frac{\left(-c^2 r + 2 m G\right) dt^2}{r} \quad (44)$$

```
> f(r, t, m) := r^2 ·dtheta^2 + r^2 ·sin(θ)^2 ·dphi^2 - c^6 r/(m^2 G^2 (-r c^2 + 2 m G)) ·dr^2
  + (-r c^2 + 2 m G)/r ·dt^2 ;
```

$$f := (r, t, m) \mapsto r^2\, d\theta^2 + r^2 \sin(\theta)^2\, d\phi^2 - c^6 r \frac{1}{m^2 G^2 \left(-r c^2 + 2 m G\right)} dr^2 + \left(-r c^2 + 2 m G\right) \frac{1}{r} dt^2 \quad (45)$$

```
> change_of_vars := {m = 2· M/Rs};
  # The change of variable help us to cancel the Energy Tensor and prove the initial equation
```

$$change_of_vars := \left\{m = \frac{2 M}{Rs}\right\} \quad (46)$$

```
> tmp := dchange(change_of_vars, [f(r, t, m)], [M], r, t, m, );
```

$$tmp := \left[r^2\, d\theta^2 + r^2 \sin(\theta)^2\, d\phi^2 - \frac{c^6 r Rs^2 dr^2}{4 M^2 G^2 \left(-c^2 r + \frac{4 M G}{Rs}\right)} + \frac{\left(-c^2 r + \frac{4 M G}{Rs}\right) dt^2}{r}\right] \quad (47)$$

```
> ds2 := r^2·dθ^2 + r^2·sin(θ)^2 dϕ^2 - c^6 r Rs^2 dr^2 / (4 M^2 G^2 (-r c^2 + 4 M G / Rs)) + (-r c^2 + 4 M G / Rs) dt^2 / r;
```

$$ds2 := r^2\,d\theta^2 + r^2\sin(\theta)^2\,d\phi^2 - \frac{c^6\,r\,Rs^2\,dr^2}{4\,M^2\,G^2\left(-c^2\,r + \frac{4\,M\,G}{Rs}\right)} + \frac{\left(-c^2\,r + \frac{4\,M\,G}{Rs}\right)dt^2}{r} \tag{48}$$

```
> Setup(coordinates = spherical, metric = ds2);
```

Systems of spacetime coordinates are: $\{X = (r, \theta, \phi, t)\}$

Coordinates: $[r, \theta, \phi, t]$. *Signature:* $(- - - +)$

$$g_{\mu,\nu} = \begin{bmatrix} -\frac{c^6\,r\,Rs^3}{16\,G^2\,M^2\left(-\frac{c^2\,r\,Rs}{4} + M\,G\right)} & 0 & 0 & 0 \\ 0 & r^2 & 0 & 0 \\ 0 & 0 & r^2\sin(\theta)^2 & 0 \\ 0 & 0 & 0 & \frac{-c^2\,r\,Rs + 4\,M\,G}{Rs\,r} \end{bmatrix}$$

$$\left[coordinatesystems = \{X\},\ metric = \left\{(1,1) = -\frac{c^6\,r\,Rs^3}{16\,G^2\,M^2\left(-\frac{c^2\,r\,Rs}{4} + M\,G\right)},\ (2,2) = r^2,\ (3,3) = r^2\sin(\theta)^2,\ (4,4) = \frac{-c^2\,r\,Rs + 4\,M\,G}{Rs\,r}\right\}\right] \tag{49}$$

```
> Ricci[scalarsdefinition];
```

$$\Phi_{00} = -\frac{R^{\mu,\nu}\,l_\mu\,l_\nu}{2},\ \Phi_{01} = -\frac{R^{\mu,\nu}\,l_\mu\,m_\nu}{2},\ \Phi_{02} = -\frac{R^{\mu,\nu}\,m_\mu\,m_\nu}{2},\ \Phi_{11} = -\frac{R^{\mu,\nu}\left(l_\mu\,n_\nu + m_\mu\,\overline{m}_\nu\right)}{4},\ \Phi_{12} = -\frac{R^{\mu,\nu}\,m_\mu\,n_\nu}{2},\ \Phi_{22} = -\frac{R^{\mu,\nu}\,n_\mu\,n_\nu}{2},\ \Lambda = \frac{R^\mu_\mu}{24} \tag{50}$$

```
> Ricci[scalars];
```

$$\Phi_{00} = 0,\ \Phi_{01} = 0,\ \Phi_{02} = 0,\ \Phi_{11} = \frac{c^4\,Rs^2 - 4\,G^2\,M^2}{4\,r^2\,c^4\,Rs^2},\ \Phi_{12} = 0,\ \Phi_{22} = 0,\ \Lambda = \frac{c^4\,Rs^2 - 4\,G^2\,M^2}{12\,r^2\,c^4\,Rs^2} \tag{51}$$

```
> Ricci[scalar];
```

$$-\frac{2\left(-c^4 Rs^2+4G^2M^2\right)}{r^2 c^4 Rs^2} \tag{52}$$

> `Ricci[scalar]`

$$-\frac{2\left(-c^4 Rs^2+4G^2M^2\right)}{r^2 c^4 Rs^2} \tag{53}$$

> `g_[ ]`

$$g_{\mu,\nu}=\begin{bmatrix} -\dfrac{c^6 r Rs^3}{16G^2M^2\left(-\dfrac{c^2 r Rs}{4}+MG\right)} & 0 & 0 & 0 \\ 0 & r^2 & 0 & 0 \\ 0 & 0 & r^2\sin(\theta)^2 & 0 \\ 0 & 0 & 0 & \dfrac{-c^2 r Rs+4MG}{Rs\, r} \end{bmatrix} \tag{54}$$

> `Λ := (c^4 Rs^2 − 4 M^2 G^2)/(12 r^2 c^4 Rs^2); Ricci[~μ, ν, matrix];`

$$\Lambda := \frac{c^4 Rs^2-4G^2M^2}{12 r^2 c^4 Rs^2}$$

$$R^{\mu}{}_{\nu}=\begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & \dfrac{c^4 Rs^2-4G^2M^2}{r^2 c^4 Rs^2} & 0 & 0 \\ 0 & 0 & \dfrac{c^4 Rs^2-4G^2M^2}{r^2 c^4 Rs^2} & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \tag{55}$$

> `1/2 ·g_[ ]·Ricci[scalar] − Ricci[~μ, ν, matrix] + Λ·g_[ ]`

`#Thus the equation for Pto generated from the Sun we make the transformation m=2·M/Rs`

$$-\frac{g_{\mu,\nu}\left(-c^4 Rs^2+4G^2M^2\right)}{r^2 c^4 Rs^2}-R^{\mu}{}_{\nu}+\frac{\left(c^4 Rs^2-4G^2M^2\right)g_{\mu,\nu}}{12 r^2 c^4 Rs^2}= \tag{56}$$

$$\left[\begin{array}{l} \dfrac{\left(-c^4\,Rs^2+4\,G^2\,M^2\right)c^2\,Rs}{16\,r\,G^2\,M^2\left(-\frac{c^2\,r\,Rs}{4}+M\,G\right)}-\dfrac{\left(c^4\,Rs^2-4\,G^2\,M^2\right)c^2\,Rs}{192\,r\,G^2\,M^2\left(-\frac{c^2\,r\,Rs}{4}+M\,G\right)} \quad \cdots \\ 0 \quad -\cdots \\ 0 \quad \cdots \\ 0 \quad \cdots \end{array}\right]$$

```
> factor( (-c^4 Rs^2 + 4 G^2 M^2) c^2 Rs / (16 r G^2 M^2 (-c^2 r Rs/4 + M G)) - (c^4 Rs^2 - 4 G^2 M^2) c^2 Rs / (192 r G^2 M^2 (-c^2 r Rs/4 + M G)) ); factor(
  - (-c^4 Rs^2 + 4 G^2 M^2)/(c^4 Rs^2) - (c^4 Rs^2 - 4 G^2 M^2)/(r^2 c^4 Rs^2) + (c^4 Rs^2 - 4 G^2 M^2)/(12 c^4 Rs^2) ); factor(
  - (-c^4 Rs^2 + 4 G^2 M^2)(-c^2 r Rs + 4 M G)/(r^3 c^4 Rs^3)
  + (c^4 Rs^2 - 4 G^2 M^2)(-c^2 r Rs + 4 M G)/(12 r^3 c^4 Rs^3) );
```

$$\frac{13\left(-c^2\,Rs+2\,MG\right)\left(c^2\,Rs+2\,MG\right)c^2\,Rs}{48\,r\,G^2\,M^2\left(-c^2\,r\,Rs+4\,MG\right)}$$

$$-\frac{\left(13\,r^2-12\right)\left(-c^2\,Rs+2\,MG\right)\left(c^2\,Rs+2\,MG\right)}{12\,r^2\,c^4\,Rs^2}$$

$$-\frac{13\left(-c^2\,Rs+2\,MG\right)\left(c^2\,Rs+2\,MG\right)\left(-c^2\,r\,Rs+4\,MG\right)}{12\,r^3\,c^4\,Rs^3} \tag{57}$$

```
> N := [[ 13 (-c^2 Rs + 2 M G)(c^2 Rs + 2 M G) c^2 Rs / (48 r M^2 G^2 (-r c^2 Rs + 4 M G)), 0, 0, 0 ],
  [ 0, -(13 r^2 - 12)(-c^2 Rs + 2 M G)(c^2 Rs + 2 M G)/(12 r^2 c^4 Rs^2), 0, 0 ],
  [ 0, 0, -(13 r^2 - 12)(-c^2 Rs + 2 M G)(c^2 Rs + 2 M G)/(12 r^2 c^4 Rs^2) · sin(θ)^2, 0 ],
  [ 0, 0, 0, -13 (-c^2 Rs + 2 M G)(c^2 Rs + 2 M G)(-r c^2 Rs + 4 M G)/(12 r^3 c^4 Rs^3) ] ];
```

$$N := \tag{58}$$

$$\begin{bmatrix} \frac{\left(-13\,c^2\,Rs+26\,MG\right)\left(c^2\,Rs+2\,MG\right)c^2\,Rs}{48\,r\,G^2\,M^2\left(-c^2\,r\,Rs+4\,MG\right)} & 0 & \cdots \\ 0 & -\frac{\left(13\,r^2-12\right)\left(-c^2\,Rs+2\,MG\right.}{12\,r^2\,c^4\,Rs^2} & \cdots \\ 0 & 0 & \cdots \\ 0 & 0 & \cdots \end{bmatrix}$$

> $a := N[1,1]; b := N[4,4]; d := N[4,4];$

$$a := \frac{\left(-13\,c^2\,Rs+26\,MG\right)\left(c^2\,Rs+2\,MG\right)c^2\,Rs}{48\,r\,G^2\,M^2\left(-c^2\,r\,Rs+4\,MG\right)}$$

$$b := -\frac{\left(-13\,c^2\,Rs+26\,MG\right)\left(c^2\,Rs+2\,MG\right)\left(-c^2\,r\,Rs+4\,MG\right)}{12\,r^3\,c^4\,Rs^3}$$

$$d := -\frac{\left(-13\,c^2\,Rs+26\,MG\right)\left(c^2\,Rs+2\,MG\right)\left(-c^2\,r\,Rs+4\,MG\right)}{12\,r^3\,c^4\,Rs^3} \quad (59)$$

> $solve(a=0);$

Warning, solve may be ignoring assumptions on the input variables.

$$\{G=G, M=M, Rs=Rs, c=0, r=r\}, \{G=G, M=M, Rs=0, c=c, r=r\}, \left\{G=\frac{c^2\,Rs}{2\,M}, M=M, Rs=Rs, c=c, r=r\right\}, \left\{G=-\frac{c^2\,Rs}{2\,M}, M=M, Rs=Rs, c=c, r=r\right\} \quad (60)$$

> $solve(b=0);$

Warning, solve may be ignoring assumptions on the input variables.

$$\left\{G=\frac{c^2\,Rs}{2\,M}, M=M, Rs=Rs, c=c, r=r\right\}, \left\{G=-\frac{c^2\,Rs}{2\,M}, M=M, Rs=Rs, c=c, r=r\right\}, \left\{G=\frac{c^2\,r\,Rs}{4\,M}, M=M, Rs=Rs, c=c, r=r\right\} \quad (61)$$

> $solve(d=0)$

Warning, solve may be ignoring assumptions on the input variables.

$$\left\{G=\frac{c^2\,Rs}{2\,M}, M=M, Rs=Rs, c=c, r=r\right\}, \left\{G=-\frac{c^2\,Rs}{2\,M}, M=M, Rs=Rs, c=c, r=r\right\}, \left\{G=\frac{c^2\,r\,Rs}{4\,M}, M=M, Rs=Rs, c=c, r=r\right\} \quad (62)$$

```
> N0 := Matrix([[0,0,0,0],[0,0,0,0],[0,0,0,0],[0,0,0,0]]);#Thus for Rs = G·2 M/c^2 we have null N0;
```

$$N0 := \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \tag{63}$$

```
> EnergyMomentum[definition] ; EnergyMomentum[ ]
```

$$T_{\mu,\nu} = \frac{G_{\mu,\nu}}{8\,\pi}$$

$$T_{\mu,\nu} = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \tag{64}$$

```
> 1/2 ·g_[ ]·Ricci[scalar]− Ricci[~μ, ν, matrix] + Λ·g_[ ] = k·T_{μ,ν}
  # Thus the equation is valid;
```

$$-\frac{g_{\mu,\nu}\left(-c^4\,Rs^2+4\,G^2\,M^2\right)}{r^2\,c^4\,Rs^2} - R^{\mu}{}_{\nu} + \frac{\left(c^4\,Rs^2-4\,G^2\,M^2\right)g_{\mu,\nu}}{12\,r^2\,c^4\,Rs^2} =$$

$$\left[\begin{array}{l} \dfrac{\left(-c^4\,Rs^2+4\,G^2\,M^2\right)c^2\,Rs}{16\,r\,G^2\,M^2\left(-\frac{c^2\,r\,Rs}{4}+MG\right)}-\dfrac{\left(c^4\,Rs^2-4\,G^2\,M^2\right)c^2\,Rs}{192\,r\,G^2\,M^2\left(-\frac{c^2\,r\,Rs}{4}+MG\right)}\ \cdots \\ 0\ -\cdots \\ 0\ \cdots \\ 0\ \cdots \end{array}\right]=$$

```
> (-c^2 Rs+2 M G) (c^2 Rs+2 M G)·[[13 c^2·Rs/(48 r M^2 G^2 (-r c^2 Rs+4 M G)), 0, 0, 0],
    [0, (13 r^2-12)/(12 r^2 c^4 Rs^2), 0, 0],
    [0, 0, (13 r^2-12)/(12 r^2 c^4 Rs^2)·sin(θ)^2, 0],
    [0, 0, 0, -13·(-r c^2 Rs+4 M G)/(12 r^3 c^4 Rs^3)]];
```

$$\left[\begin{array}{ll} \dfrac{13\left(-c^2\,Rs+2\,MG\right)\left(c^2\,Rs+2\,MG\right)c^2\,Rs}{48\,r\,G^2\,M^2\left(-c^2\,r\,Rs+4\,MG\right)} & 0\ \cdots \\ 0 & \dfrac{\left(13\,r^2-12\right)\left(-c^2\,Rs+2\,MG\right)(}{12\,r^2\,c^4\,Rs^2}\cdots \\ 0 & 0\ \cdots \\ 0 & 0\ \cdots \end{array}\right] \quad (66)$$

```
>
```

```
> series(13 c^2·Rs/(48 r M^2 G^2 (-r c^2 Rs+4 M G)), r); series((13 r^2-12)/(12 r^2 c^4 Rs^2), r); series(
    -13·(-r c^2 Rs+4 M G)/(12 r^3 c^4 Rs^3), r);
```

$$\frac{13\,c^2\,Rs}{192\,G^3\,M^3}\,r^{-1}+\frac{13\,c^4\,Rs^2}{768\,G^4\,M^4}+\frac{13}{3072}\,\frac{c^6\,Rs^3}{G^5\,M^5}\,r+\frac{13}{12288}\,\frac{c^8\,Rs^4}{G^6\,M^6}\,r^2+\frac{13}{49152}\,\frac{c^{10}\,Rs^5}{G^7\,M^7}\,r^3$$
$$+\frac{13}{196608}\,\frac{c^{12}\,Rs^6}{G^8\,M^8}\,r^4+\frac{13}{786432}\,\frac{c^{14}\,Rs^7}{G^9\,M^9}\,r^5+O\left(r^6\right)$$

$$-\frac{1}{c^4\,Rs^2}\,r^{-2}+\frac{13}{12\,c^4\,Rs^2}$$

$$-\frac{13\,MG}{3\,c^4\,Rs^3}\,r^{-3}+\frac{13}{12}\,\frac{1}{c^2\,Rs^2}\,r^{-2} \tag{67}$$

> `solve( - Rs· c^2 + 2· M·G);`

$$\left\{G=\frac{c^2\,Rs}{2\,M},\ M=M,\ Rs=Rs,\ c=c\right\} \tag{68}$$

> `c := 299792458; m := 1.9885·10^30; G := 6.6743·10^-11 ;`

$$c := 299792458$$

$$m := 1.988500000\times 10^{30}$$

$$G := 6.674300000\times 10^{-11} \tag{69}$$

> `Rs := evalf( 2· m·G / c^2 );`

$$Rs := 2953.383938 \tag{70}$$

> `M := c^2·Rs / 2· G ;`

$$M := 1.988500000\times 10^{30} \tag{71}$$

> `-Rs ·c^2 + 2· M· G; Rs·c^2; 2·M·G;`

$$1.\times 10^{11}$$

$$2.654369109\times 10^{20}$$

$$2.654369110\times 10^{20} \tag{72}$$

> `Rs^2·c^4; 4·M^2·G^2; series( - c^6 r Rs^3 / ( 16 G^2 M^2 ( - c^2 r Rs / 4 + M G ) ), r ); series( - 4 M^2 G^2 r / ( Rs c^2 ( -c^2 r Rs + 4 G M ) ), r );`

$$7.045675367\times 10^{40}$$

$$7.045675372\times 10^{40}$$

$$(-0.4999999994)\,r-0.2499999996\,r^2-0.1249999997\,r^3-0.06249999983\,r^4-0.03124999991\,r^5+O\left(r^6\right)$$

$$(-0.5000000001)\,r-0.2499999999\,r^2-0.1249999999\,r^3-0.06249999992\,r^4-0.03124999994\,r^5+O\left(r^6\right) \tag{73}$$

>